

Ex.No:05

Create custom views and animations using Android's Canvas and Animator classes. Implement transitions and interactive animations in iOS using Swift and UIView animation APIs.

STEP 1: Create New Project (Kotlin)

1. Open Android Studio
2. Click **New Project**
3. Select **Empty Activity**
4. Click **Next**
5. Enter:
 - **Name:** CustomAnimationApp
 - **Language:** Kotlin ✓
 - **Minimum SDK:** API 21+
6. Click **Finish**
7. Wait for build to complete

STEP 2: Create Custom View Class

1. Go to:
app > java > your_package_name
2. Right-click → **New** → **Kotlin Class/File**
3. Name it: CustomView
4. Select **Class**
5. Paste this code:

```
import android.animation.ValueAnimator
import android.content.Context
import android.graphics.Canvas
import android.graphics.Paint
import android.view.View

class CustomView(context: Context) : View(context) {
```

```

private val paint = Paint()
private var radius = 50f

init {
    paint.color = 0xFF6200EE.toInt()
    startAnimation()
}

override fun onDraw(canvas: Canvas) {
    super.onDraw(canvas)
    canvas.drawCircle(width / 2f, height / 2f, radius, paint)
}

private fun startAnimation() {
    val animator = ValueAnimator.ofFloat(50f, 200f)
    animator.duration = 2000
    animator.repeatMode = ValueAnimator.REVERSE
    animator.repeatCount = ValueAnimator.INFINITE

    animator.addUpdateListener {
        radius = it.animatedValue as Float
        invalidate()
    }

    animator.start()
}
}

```

STEP 3:

1. app → src → main → java → com.example.customanimationapp → MainActivity.kt

2. Copy and paste this code inside MainActivity.kt:

```

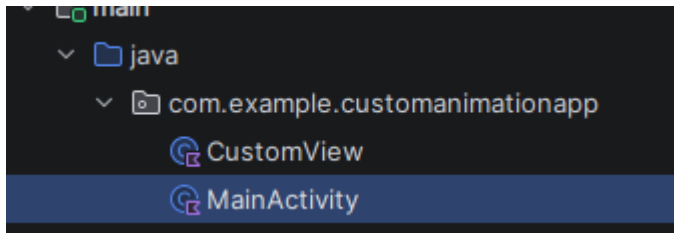
package com.example.customanimationapp

import android.os.Bundle
import androidx.appcompat.app.AppCompatActivity

class MainActivity : AppCompatActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(CustomView(this))
    }
}

```

NOTE: Both the customView and MainActivity should be in the same package



STEP 4:

Copy and paste this code in AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">

    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="CustomAnimationApp"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.CustomAnimationApp">

        <activity
            android:name=".MainActivity"
            android:exported="true">

            <intent-filter>
                <action android:name="android.intent.action.MAIN" />
                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>

        </activity>

    </application>

</manifest>
```

STEP 5:

Run the code

OUTPUT:

